For $j \geq 2$:

$$v_j^{(2)} = v_j^{(1)} - \eta c' \frac{\partial \mathcal{L}^0}{\partial v_j} + \frac{\eta\,(\rho - c')}{t} \frac{(t-j+1)(t-j+2)}{2} \mathbb{E}_{a,b}\Bigg[2\frac{(a-1)(b-1)\left((b-1)\left(a-\frac{1}{2}\right)+(a-1)\left(b-\frac{1}{2}\right)\right)}{(\lambda-1)^3}\lambda^{j-1} + \frac{(b-a)\left((b-1)^2\left(a-\frac{1}{2}\right)-(a-1)^2\left(b-\frac{1}{2}\right)\right)}{(\lambda-1)^3}\Bigg] + O(1), \tag{A.48}$$

where for all j it is:

$$v_j^{(1)} - \eta c' \frac{\partial \mathcal{L}^0}{\partial v_j} = c + \frac{(c+c')\eta}{t} \frac{(t-j+1)(t-j+2)}{2} \mathbb{E}_{a,b}\left[\frac{(a-1)^2+(b-1)^2}{(\lambda-1)^2} - \frac{1}{2}\right] + O(1). \tag{A.49}$$

We now show that $v_2^{(2)} > v_1^{(2)}$ in the large t regime. Observe that the term of (A.49) is the same for $j = 1, 2$. We evaluate the expectation of eq. (A.47) that corresponds to $j = 1$:

$$\mathbb{E}_{a,b}\left[\frac{(b-1)^2(a-\frac{1}{2})+(a-1)^2(b-\frac{1}{2})}{(\lambda-1)^2}\right] = \frac{1-\ln 2}{3} \tag{A.50}$$

and the expectation of (A.48), leveraging symmetry of a, b in the expression, that corresponds to $j = 2$:

$$4\mathbb{E}_{a,b}\left[\frac{(a-1)(b-1)^2(a-\frac{1}{2})}{(a+b-2)^3}(a+b-1)\right] + 2\mathbb{E}_{a,b}\left[\frac{(b-a)(b-1)^2(a-\frac{1}{2})}{(a+b-2)^3}\right] = \frac{7-10\ln 2}{2} + \frac{5-8\ln 2}{6}. \tag{A.51}$$

The latter term is larger, so, when $t \to \infty$, it is $v_2^{(2)} > v_1^{(2)}$. Finally, observe that the expectation $\mathbb{E}_{a,b}\left[\frac{(a-1)(b-1)^2(a-\frac{1}{2})}{(a+b-2)^3}(a+b-1)^{j-1}\right]$ is negative for j odd and decreasing otherwise, which shows that $v_2^{(2)} > v_j^{(2)}, j > 2$. So, we showed that the 2nd coordinate grows larger than the rest after the 2nd step.

□

With this calculation at hand, we can ask questions about the optimization trajectory and how that would change if we alter the data distribution. In particular, in the following Corollary, we answer the question of how the gradient would change when $a = b$, i.e., when the transition matrix is

$$\mathcal{P} = \begin{pmatrix} a & 1-a \\ 1-a & a \end{pmatrix}, \tag{A.52}$$

with $a \sim Unif([0,1])$. Notice that in the case there is not a unigrams solution (or in other words the unigrams solution is as good as the uniformly at random solution).

Corollary A.2. *When $a = b$, it is*

$$\begin{aligned} W_k^{(1)} &= \begin{pmatrix} c & c \\ c & c \end{pmatrix} + c\eta\left[O(t)\begin{pmatrix} \infty & 0 \\ 0 & \infty \end{pmatrix} + O(1)\right], \\ v_j^{(1)} &= c - \frac{c\eta}{2t}(t-j+1) + O(1/t), j \in [t]. \end{aligned} \tag{A.53}$$

Proof. We first compute the gradient with respect to W_k, by setting $a = b$ in eq. (A.27),(A.28). For $m, l = 1, 1$, we have:

$$\left(W_k^{(1)}\right)_{1,1} = c + \frac{\eta c}{t}\mathbb{E}_{a\sim Unif([0,1])}\left[\frac{(a-1)(a-\frac{1}{2})}{(2a-2)^3}\left[\frac{2(a-1)^2}{(2a-2)^2}(II) + \frac{2(a-1)^2}{2a-2}(III)\right]\right], \tag{A.54}$$

where $(II), (III)$ are defined in eq. (A.31). Notice that the (IV) with the cubic, $O(t^3)$, dependence disappeared, so the leading order terms are $O(t^2)$. By substituting the leading order terms from $(II), (III)$, we get:

$$\begin{aligned}\left(W_k^{(1)}\right)_{1,1} &= c + \frac{\eta c}{t}\mathbb{E}_{a\sim Unif([0,1])}\left[\frac{(a-1)(a-\frac{1}{2})}{(2a-2)^3}\left[\frac{2(a-1)^2}{(2a-2)^2}\frac{2-2a}{2} - \frac{2(a-1)^2}{2a-2}\frac{1}{2}\right]\right]O(t^2) \\ &= c + \eta c\mathbb{E}_{a\sim Unif([0,1])}\left[-\frac{(a-\frac{1}{2})}{8(a-1)}\right]O(t) \to \infty.\end{aligned} \tag{A.55}$$

For $m, l = 1, 2$, the $O(t^3)$ terms similarly disappear, and we have:

$$\begin{aligned}\left(W_k^{(1)}\right)_{1,2} &= c + \frac{\eta c}{t}\mathbb{E}_{a\sim Unif([0,1])}\left[\frac{(a-1)(a-\frac{1}{2})}{(2a-2)^3}\left[\frac{2(a-1)^2}{(2a-2)^2}(II) - \frac{2(a-1)^2}{2a-2}(III)\right]\right] \\ &= c + \frac{\eta c}{t}\mathbb{E}_{a\sim Unif([0,1])}\left[\frac{(a-1)(a-\frac{1}{2})}{(2a-2)^3}\left[\frac{2(a-1)^2}{(2a-2)^2}\frac{2-2a}{2} + \frac{2(a-1)^2}{2a-2}\frac{1}{2}\right]\right]O(t^2) \\ &= c.\end{aligned} \tag{A.56}$$

Now, for the v vector, we observe that when we set $a = b$ in eq. (A.36), the leading term disappears and its expression simplifies to:

$$\begin{aligned}v_j^{(1)} &= c - \frac{c\eta}{t}\mathbb{E}_{a\sim Unif([0,1])}\left[2(2a-2)\frac{(a-1)^2}{(2a-2)^3}(t-j+1)\right] + O(1/t) \\ &= c - \frac{c\eta}{2t}(t-j+1) + O(1/t).\end{aligned} \tag{A.57}$$

□

The previous calculations reveal that there is less "signal" towards the unigrams solution when $a = b$.

Finally, we provide our main technical result that demonstrates that two steps of stochastic gradient descent lead the model to the bigrams solution.

Proposition A.3. *Suppose we initialize the model of eq.* (A.10) *with* $W_k^{(0)} = w_{init}11^T$ *and* $v = v_{init}1^T$. *Consider a two step procedure, where we first train with gradient descent on the original distribution* $a \sim \mathrm{Unif}(0,1), b \sim \mathrm{Unif}(0,1)$, *and then at the second step with* $a = b \sim \mathrm{Unif}([\frac{1}{2}-\epsilon, \frac{1}{2}+\epsilon])$, $\epsilon \in (0, 1/2)$. *Let* $\eta_1 = O\left(\frac{1}{t^2}\right)$ *and* $\eta_2 = O\left(\frac{1}{t}\right)$ *be the learning rates for the 1st and 2nd step of gradient descent, and let* $\lambda_{wd} = 1$ *be the weight decay coefficient used in the second step. Then, after 2 steps of gradient descent, for* $t \to \infty$, *it is:*

$$f(e)_{p,i} = O(\epsilon^4)\sum_{t'=1}^{p}\mathbb{1}\{x_{t'} = i\}\mathbb{1}\{x_p = x_{t'-1}\} + O(\epsilon^6). \tag{A.58}$$

Proof. For the first step, we have from Lemma 1:

$$W_k^{(1)} = w_{init}11^T + v_{init}\eta_1\begin{pmatrix}B & A\\ A & B\end{pmatrix}O(t^2) + v_{init}\eta_1 O(t). \tag{A.59}$$

For $\eta_1 = O\left(\frac{1}{t^2}\right)$, then

$$W_k^{(1)} = w_{init}11^T + \begin{pmatrix}B & A\\ A & B\end{pmatrix}O(1) + O(1/t). \tag{A.60}$$

Recall, from Lemma 1, that we have $v_j^{(1)} = v_{init} + \eta_1 w_{init}O(t)$ for all $j \in [t]$, so, for $\eta_1 = O\left(\frac{1}{t^2}\right)$, it will be $v_j^{(1)} = v_{init}$ for all $j \in [t]$.

Assume now that at the second step we train with weight decay $\lambda_{wd} = 1$ and the data distribution changes to $a = b \sim \text{Unif}([\frac{1}{2} - \epsilon, \frac{1}{2} + \epsilon])$. Then, from eq. (A.55)[2] the update of the 2nd layer would be:

$$\begin{aligned} W_k^{(2)} &= (1 - \lambda_{wd}) W_k^{(1)} - \eta_2 \nabla \mathcal{L} \\ &= v_{init} \eta_2 \begin{pmatrix} \int_{\frac{1}{2}-\epsilon}^{\frac{1}{2}+\epsilon} \frac{a-\frac{1}{2}}{(a-1)} & 0 \\ 0 & \int_{\frac{1}{2}-\epsilon}^{\frac{1}{2}+\epsilon} \frac{a-\frac{1}{2}}{(a-1)} \end{pmatrix} O(t) + O(1) \\ &= v_{init} \eta_2 \begin{pmatrix} 2\epsilon + \frac{1}{2} \ln \left(\frac{\frac{1}{2}-\epsilon}{\frac{1}{2}+\epsilon} \right) & 0 \\ 0 & 2\epsilon + \frac{1}{2} \ln \left(\frac{\frac{1}{2}-\epsilon}{\frac{1}{2}+\epsilon} \right) \end{pmatrix} O(t) + O(1). \end{aligned} \tag{A.61}$$

For the update of v we have:

$$\begin{aligned} v_j^{(2)} &= (1 - \lambda_{wd}) v_j^{(1)} - \frac{\partial \mathcal{L}}{\partial v_j} \\ &= \eta_2 A \frac{\partial \mathcal{L}_{const}}{\partial v_j} + \eta_2 3A \frac{\partial \mathcal{L}_{diag}}{\partial v_j} \\ &= A \frac{\eta_2 \rho}{t} \left(\frac{(t-j+1)(t-j+2)}{2} \left(\frac{(a-1)^2 + (b-1)^2}{(\lambda - 1)^2} - \frac{1}{2} \right) + O(t) \right) + 3A\eta_2 \frac{\partial \mathcal{L}_{diag}}{\partial v_j}. \end{aligned} \tag{A.62}$$

For $a = b$, it is only the $O(t)$ part inside the parenthesis that survives from the second term (see eq. (A.57)) and by substituting the diagonal gradient from eqs. (A.47),(A.48) when $a = b$, we get for all $j \in [t]$:

$$v_j = \eta_2 \rho A O(1) + 3A \frac{\eta_2}{t} \frac{(t-j+1)(t-j+2)}{2} 2\mathbb{E}_{a \sim Unif([1/2-\epsilon, 1/2+\epsilon])} \left[\frac{a - 1/2}{4} (2a-1)^{j-1} \right] + 3A \frac{\eta_2}{t} O(t). \tag{A.63}$$

We calculate the expectation:

$$\mathbb{E}_{a \sim Unif([1/2-\epsilon, 1/2+\epsilon])} \left[\frac{a - 1/2}{4} (2a-1)^{j-1} \right] = 0, \tag{A.64}$$

when $j = 1$. For $j \geq 2$:

$$\begin{aligned} \mathbb{E}_{a \sim Unif([1/2-\epsilon, 1/2+\epsilon])} \left[(a - 1/2)(2a-1)^{j-1} \right] &= \int_{1/2-\epsilon}^{1/2+\epsilon} (a - 1/2)(2a-1)^{j-1} \\ &= \frac{1}{2j} \left[(2a-1)^j (a - \frac{1}{2}) \right]_{1/2-\epsilon}^{1/2+\epsilon} - \frac{1}{2j} \int_{1/2-\epsilon}^{1/2+\epsilon} (2a-1)^j da \\ &= \frac{1}{2j} \epsilon \left((2\epsilon)^j + (-2\epsilon)^j \right) - \frac{1}{4j(j+1)} \left((2\epsilon)^{j+1} - (-2\epsilon)^{j+1} \right). \end{aligned} \tag{A.65}$$

For $j = 2k+1$, this equals to 0. For $j = 2k$, it is equal to $\frac{4^k}{2k+1} \epsilon^{2k+1}$.

Thus, by setting $\eta_2 = O\left(\frac{1}{t}\right)$, we get:

$$W_k^{(2)} = O(\epsilon) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + O\left(\frac{1}{t}\right). \tag{A.66}$$

$$v_j = v_{init} + O\left(\frac{(t-j+1)(t-j+2)}{t^2} \right) \sum_{s=1, s=2k}^{t} \delta(s-j) \epsilon^{j+1} + O\left(\frac{1}{t}\right), \tag{A.67}$$

[2]That equation refers to the first step of gradient descent for $a = b$, but notice that since the model is linear in W_k, and in this case $v_j^{(1)} = v_{init}$, it characterizes the 2nd step too.